E · TOTEM

GENERAL ARRANGEMENT — COMMUNITY BOARD

POLE-MOUNTED PUBLIC E-INK NOTICEBOARD + MESH NODE

AF-E-101

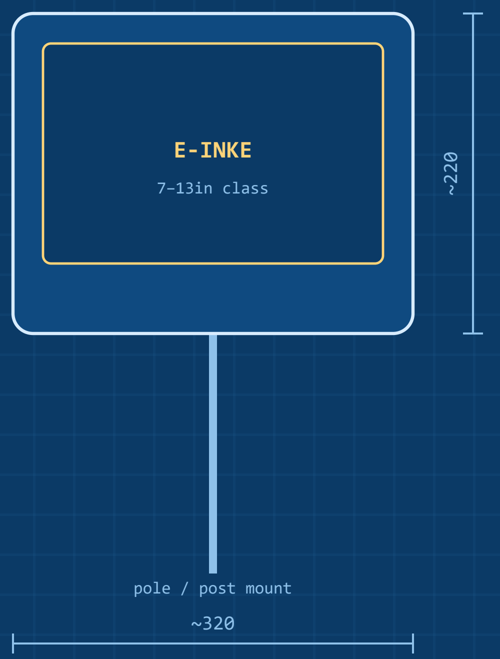
REV A · 2026-11-12

PROTOTYPE

WHAT THIS NODE IS

- The town square in a box: a public E-Ink board that is ALSO a mesh node — it doesn't just display, it routes.
- Emergency alerts, community notices, local information, agency bulletins — readable at arm's length in sunlight.
- Zero power to display: E-Ink holds an image indefinitely on stored charge — the board works in a blackout.
- The community noticeboard is familiar, trusted rural technology. Nobody fears a sign.

ENVELOPE — POLE-MOUNTED BOARD



enclosure ~320×220×100 mm · pole 3 m class · total ~8 kg

CONTENT PIPELINE

AGENCY / COMMUNITY — CMS publishes alerts →
MESH NETWORK — signed envelopes · store-forward →
TOTEM NODE — LoRa RX → render → E-Ink →
PUBLIC DISPLAY — sunlight-readable · zero idle power

USE CASES

Use	Detail
Emergency alerts	Wildfire, evacuation, Amber-style, weather — pushed mesh-wide in minutes
Community notices	Council, school, clinic, events — the digital general-store board
Agency bulletins	Province, RCMP, health unit — one board, many authors, signed content
Local info	Road conditions, water advisories, harvest deadlines, powwow/community events
Mesh status	Optional: node shows mesh health — coverage is visible infrastructure

WHY IT'S UNIQUE

- Only public noticeboard that IS a network node — it routes, repeats, and stores, not just displays.
- Signed content: every notice carries publisher identity — no anonymous flyers, no defacement.
- Offline by design: works when grid AND cell network are both down (the blackout case).
- EN/FR content pipeline; community controls what appears — sovereignty in public space.

ARCTICCHAIN · RURAL NODE FLEET

PROPRIETARY & CONFIDENTIAL — EMEBET INC.

DRAWN: Surafel Negussie — Planner / Maker

SOLE PROPRIETARY OWNER

Emebet Inc. · emebetos.ca

registered Canadian company — Edmonton, AB

DWG AF-E-101

REV A

SCALE NTS · SHEET 1/3

DESIGNED FOR RURAL & REMOTE CANADA — -45..+70 °C · IP65 · SOLAR-FIRST · COMMUNITY-OWNED

E · TOTEM

MECHANICAL & INSTALLATION

ENCLOSURE, POLE MOUNT, DISPLAY WINDOW

AF-E-102

REV A · 2026-11-12

PROTOTYPE

ENCLOSURE SPEC

Parameter	Value	Parameter	Value
[illegible] hose-down)			
[illegible] no PVC fiberglass box			
[illegible] gasketed			
[illegible]			
[illegible] plate			
[illegible] fasteners			

BOX CANDIDATES (ORDERABLE)

Option	Notes
[illegible] ×220×100	
[illegible]	
[illegible]	

INSTALLATION

- Two-person install, ~30 min: mount pole (post-hole or U-bolt to existing pole), hang box, connect solar leads.
- Orientation: display faces road/foot-traffic; antenna port top; solar panel south-facing tilt.
- No concrete required — expandable pole anchor or existing utility pole U-bolt.
- Service: front latch access to SD card + battery; field-swappable in <10 min.

DISPLAY WINDOW DETAIL

- Window: 3 mm optically clear PET film or AC sheet, gasketed to box lip.
- E-Ink sits 2–3 mm behind window (anti-static gap).
- Sun hood: 40 mm visor reduces washout + sheds water/ice.
- Optional: anti-graffiti laminate film (sacrificial, field-replaceable).

WEATHER & ENVIRONMENT

- -45..+70 °C: polycarbonate box + wide-temp components; E-Ink loves cold.
- Ice loading: sloped top surfaces shed ice/snow; display window has drip edge.
- Condensation: breather vent or desiccant pack; check annually.
- Vandalism: no sharp edges, tamper-evident seal; content is signed so defacement is detectable.

POLE MOUNT DETAIL

- Two stainless U-bolts through box rear flange, backing plate spreads load.
- Box hangs on rear hinge — swings down for service access.
- Cable entry: bottom grommet, drip loop; solar cable armored or conduit sleeve.
- Wind: box ~8 kg on 3 m pole — check pole class for 150 km/h gust site.

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REV A

SCALE NTS · SHEET 2/3

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ELECTRICAL, CONTENT & SOFTWARE

BOM + POWER + CMS PIPELINE

AF-E-103

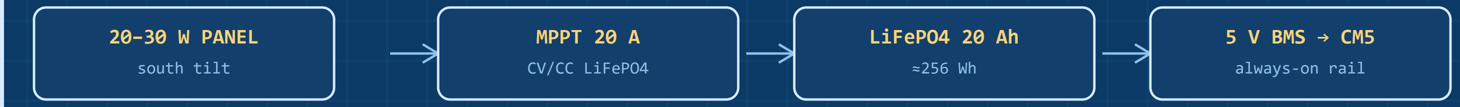
REV A · 2026-11-12

PROTOTYPE

BILL OF MATERIALS (PER TOTEM)

#	Part	Supplier class	Approx CAD	Qty
[illegible] Renesas L1 — fleet common)				
[illegible] Paper HAT (SPI)				
[illegible] 915 MHz				
[illegible] optional backhaul)				
[illegible] panel + MPPT 20 A				
[illegible] battery supplier				
[illegible]×220×100				
[illegible]window film + gasket				

POWER SYSTEM



20–30 W PANEL
south tilt

MPPT 20 A
CV/CC LiFePO4

LiFePO4 20 Ah
≈256 Wh

5 V BMS → CM5
always-on rail

Autonomy: ~256 Wh ÷ ~2–4 W avg ≈ 3–5 days no-sun; winter derate → oversized panel is the plan.

CONTENT PIPELINE — SIGNED PUBLICATIONS

- Publishers sign content (Ed25519) — community/agency keys.
- Mesh floods signed notice envelopes; node verifies before render.
- Display rotation: alerts > notices > info; expiry timestamps auto-clear.
- Rate-limit + dedup: one publisher can't spam the board.
- Offline queue: notices arrive whenever a path opens — order preserved.
- Moderation: community admin key can pin/hide categories (not delete history).

SOFTWARE

- Same arcticchain_agent.py base; Totem adds: content renderer → framebuffer → e-Paper driver.
- CMS side: simple signed-publish tool (repo: poc/ or later arcticchain-os).
- Node roles: display + repeater + optional gateway (LTE/Wi-Fi backhaul when present).

FIELD ACCEPTANCE

- Notice pushed from 10 km away renders on Totem within 60 s (mesh path).
- Board legible in full sun at 3 m; zero power draw between updates.
- Survives overnight -30 °C; resumes display + comms at dawn.
- Tamper test: locked box, signed content undetectable-defacement alarm.

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REV A

SCALE NTS · SHEET 3/3

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